

Track 4 · Shopping Copilot · Astrid CLI Engine · Zero Token Cost

ShopPilot

Offline-first multi-turn shopping copilot

TechnicalScore: 0.907

Hit@10: 97.5%

MRR: 0.858

MTTC: 2.88 turns

github.com/algorathem/techjam2026-shopping-copilot

Why Static Keyword Retrieval Breaks in Conversational Search

Single-turn search assumes fixed intent. Real shopping sessions suffer from four core failure modes:

01

Vague queries

"something nice for summer" floods the returned results

02

Semantic Collisions

"dress sandals" pulling dresses instead of footwear

03

Mind-change

outdated saved contexts

04

Turn Fatigue

miss after turn 10 → cart abandonment

Objective Function

Find the hidden parent_asin within a 50k frozen catalog under the official formula:

$$\text{TechnicalScore} = 0.50 \cdot \text{Hit@10} + 0.30 \cdot \text{MRR} + 0.20 \cdot \left(\frac{11 - \text{MTTC}}{10} \right)$$

ShopPilot Engine: Headless, Fast, and Network-Free

Every respond() turn returns a natural message, one structured ask_attribute, and Top-10 ASINs in < 15ms.



TECHJAM 2026 · TRACK 4

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Hybrid FTS + dense

Stateful DST

Clarify ≤ 10 turns

Tech 0.907

Demo UI: Astrid CLI · github.com/algorathem/techjam2026-shopping-copilot

Public 200 · Hit@10 0.975 · MTTC 2.88 · tokens 0

Public 200 — weak BM25 vs ShopPilot

Official local_evaluator · SHOPPILOT_DENSE=hash · deterministic seeds

TechnicalScore

0.907

from 0.107 · ~8.5×

Hit@10

0.975

from 0.125

MRR

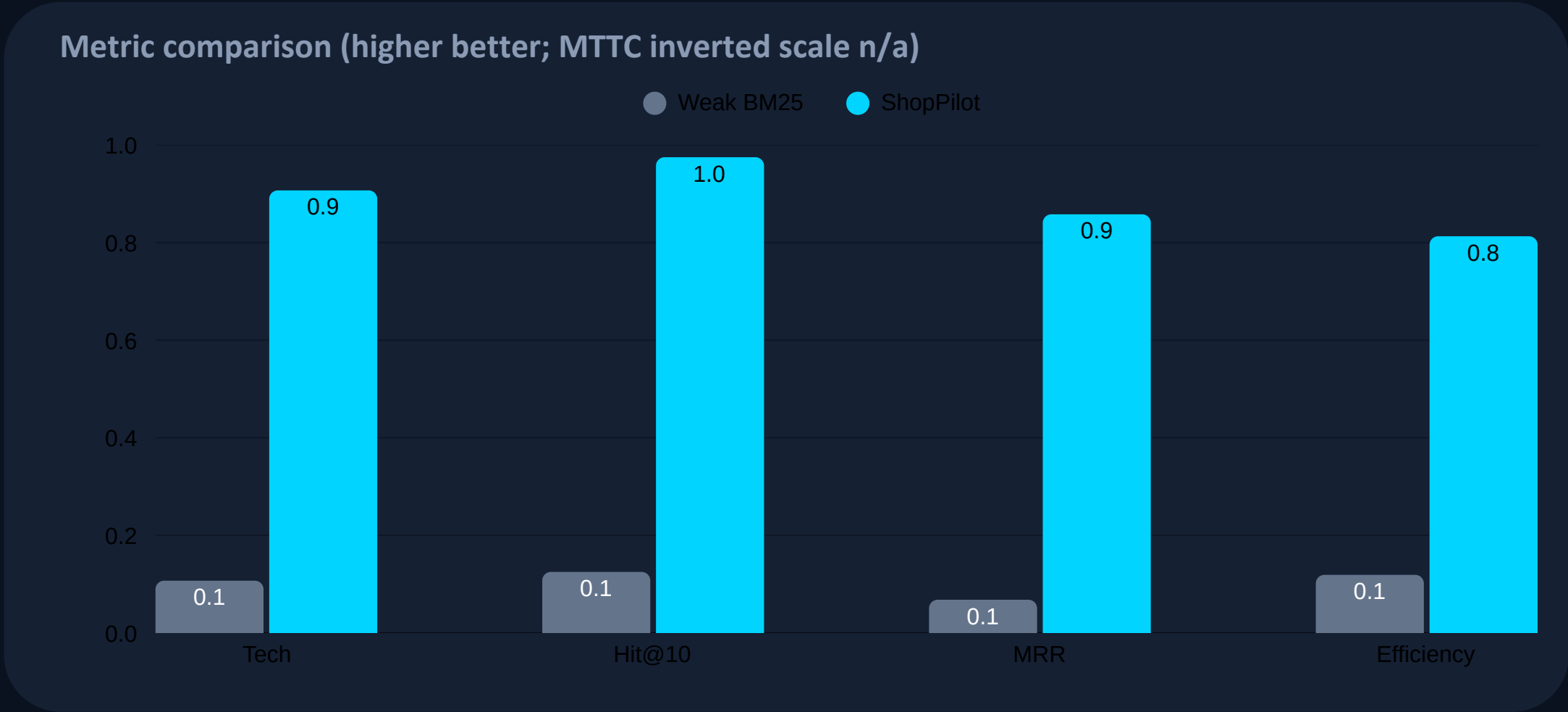
0.858

from 0.068

MTTC

2.88

from 9.81 turns



What moved the needle

- Override hygiene (soft-only wipe)
- Hybrid dense hash lane
- Family + audience routing
- other-first clarify policy
- Multi-slot freeform atomize
- Cold-start profile/rating priors

Each kept only if Tech ≥ floor

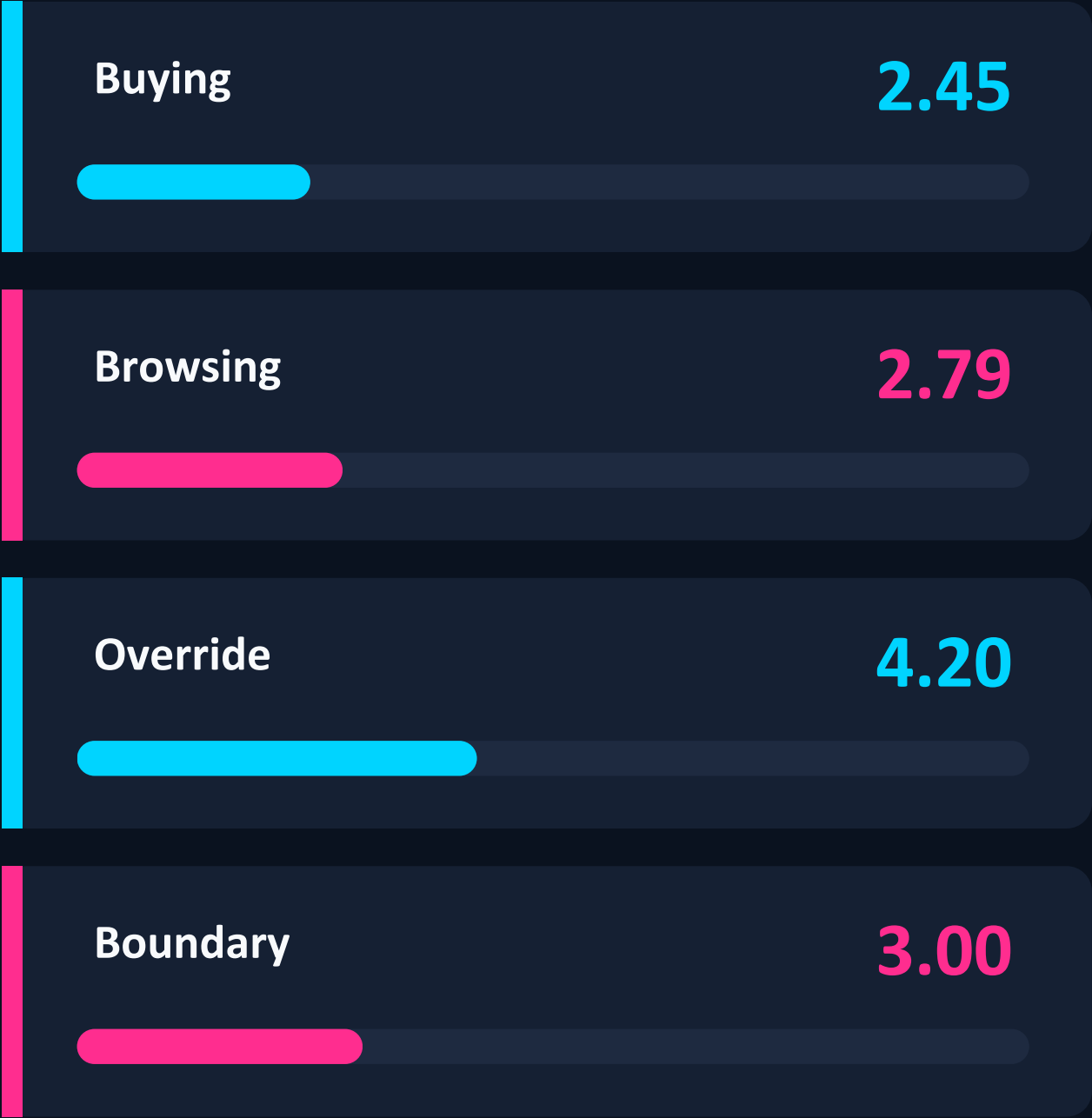
Tokens default path: 0

Hit@10 holds across buying, browse, override, boundary

Override pays more MTTC (mind-change cost) but still Hit 0.967



MTTC (turns to first hit)



Astrid CLI — live multi-turn

```
python3 cli_chat.py --dense hash · /new /state /quit
```



astrid — dense=hash

```
$ python3 cli_chat.py --dense hash
```

Astrid · quiet clarity for every aisle

You · shoes for my son

Astrid: Got it (footwear; boys). Any color...?

↳ ask · other Top-3 footwear titles

You · Actually forget sneakers, dress shoes size 9

Astrid: Updated (footwear; size 9). Color?

↳ override wiped soft prefs · kept size path

/state → family=footwear size=9 asked={other}

Vague → clarify

dress + exploring → other → plus size locks size
(no re-ask)

Family lock

"dress sandals" → footwear, not garment dresses

Intent override

running shoes → dress shoes; size 9 persists

Kit metrics → merchant outcomes

Honest framing: offline simulation ≠ live GMV A/B — same job-to-be-done as conversational commerce.

Hit@10	0.975	Findability — target ASIN in top 10
MRR	0.858	Rank quality — earlier positions win
MTTC	2.88	Cost / cognitive load — fewer refine loops
Tokens	0	Offline default — no paid LLM on score path

THANKS · QUESTIONS

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Astrid CLI · Tech 0.907 · Hit@10 0.975 · MTTC 2.88

Live demo ready

Reproduce: local_evaluator

Track 4 · CSJ 50k